

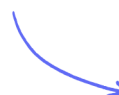
Multiple Choice Questions

Energetics

Enthalpy Level Diagrams / Enthalpy Change - Definitions / Using Calorimetry /
Using Hess Cycles / Bond Enthalpies

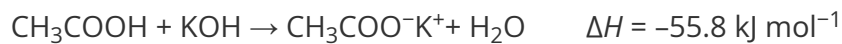
Easy (5 questions)	/5
Medium (5 questions)	/5
Total Marks	/10

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Easy Questions

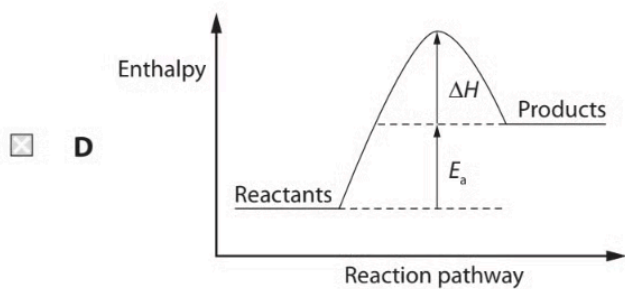
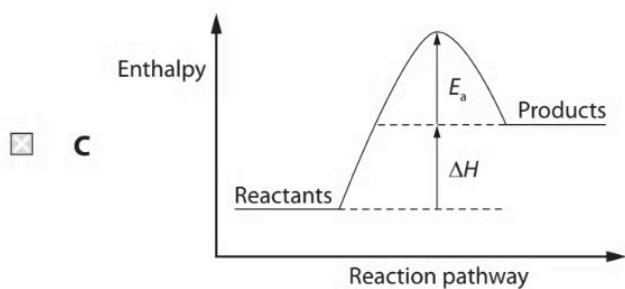
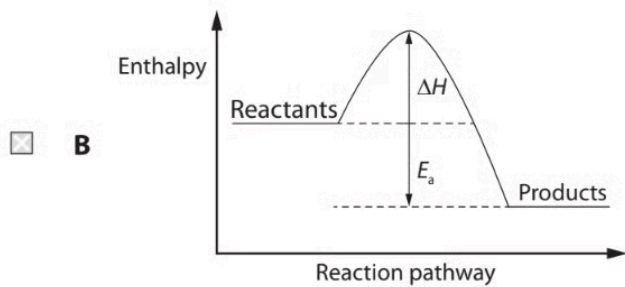
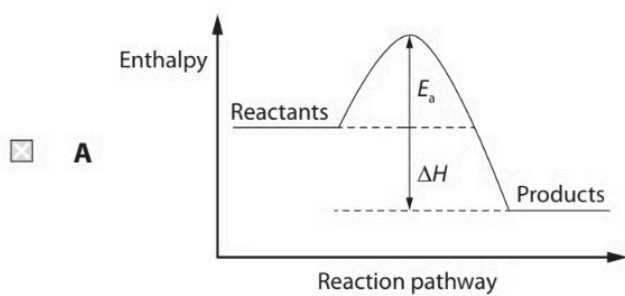
1 Which are correct for the reaction shown?



		Type of reaction	Type of enthalpy change
☐	A	endothermic	formation
☐	B	endothermic	neutralisation
☐	C	exothermic	formation
☐	D	exothermic	neutralisation

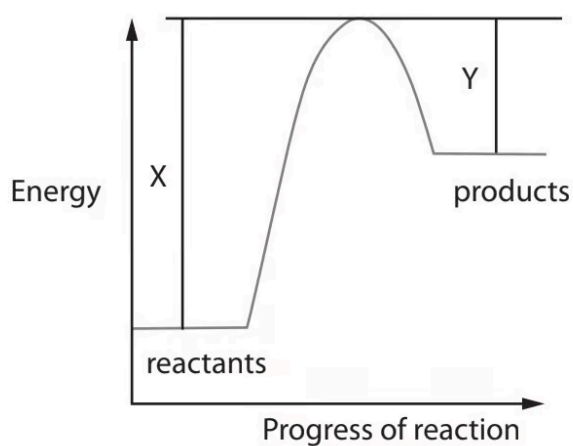
(1 mark)

2 Which is the correctly labelled reaction profile for an exothermic reaction?



(1 mark)

3 The energy profile for a reaction is shown.



What is the minimum energy needed for this reaction to occur?

- A. X
- B. Y
- C. $X - Y$
- D. $X + Y$

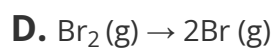
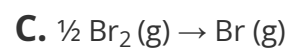
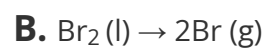
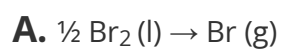
(1 mark)

4 Which equation does **not** represent a standard enthalpy change of atomisation?

- A. $\text{Mg (s)} \rightarrow \text{Mg (g)}$
- B. $\text{Cl}_2 \text{ (g)} \rightarrow 2\text{Cl (g)}$
- C. $\frac{1}{2}\text{O}_2 \text{ (g)} \rightarrow \text{O (g)}$
- D. $\text{Hg (l)} \rightarrow \text{Hg (g)}$

(1 mark)

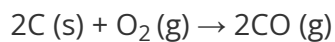
5 Which equation represents the standard enthalpy change of atomisation, $\Delta_{\text{at}}H$, of bromine?



(1 mark)

Medium Questions

- 1 The equation for a reaction is



Which is the correct symbol for the enthalpy change for this reaction?

- A. $\Delta_{\text{at}}H$
- B. $\Delta_{\text{c}}H$
- C. $\Delta_{\text{f}}H$
- D. $\Delta_{\text{r}}H$

(1 mark)

- 2 5.20 g of sodium hydrogencarbonate is added to an excess of acid.

The temperature increases and the energy change is calculated to be 1030 J.

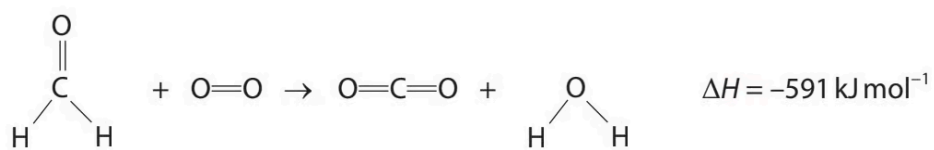
What is the enthalpy change per mole of sodium hydrogencarbonate?

[M_{r} $\text{NaHCO}_3 = 84.0$]

- A. $-12.3 \text{ kJ mol}^{-1}$
- B. $-16.6 \text{ kJ mol}^{-1}$
- C. $-63.8 \text{ kJ mol}^{-1}$
- D. $-16600 \text{ kJ mol}^{-1}$

(1 mark)

3 The equation for the complete combustion of methanal is shown.



Some bond enthalpy data are shown.

Bond	Bond enthalpy / kJ mol^{-1}
C—H	413
O = O	498
C = O in CO_2	805
O—H	464

What is the C = O bond enthalpy in methanal?

- A. 623 kJ mol^{-1}
- B. 678 kJ mol^{-1}
- C. 805 kJ mol^{-1}
- D. 1036 kJ mol^{-1}

(1 mark)

4 The table shows the amount of energy released per gram when some alkanes are burned in excess oxygen under standard conditions.

Alkane	Energy released / kJ g^{-1}
methane	55.6
ethane	52.0
propane	50.4
butane	49.6

Which alkane has a standard enthalpy change of combustion of $-2877 \text{ kJ mol}^{-1}$?

- A.** methane
- B.** ethane
- C.** propane
- D.** butane

(1 mark)

5 The enthalpy changes for two reactions are shown.



What is the enthalpy change for the dissociation of $\text{CH}_3\text{COOH}(\text{aq})$ into $\text{CH}_3\text{COO}^-(\text{aq})$ and $\text{H}^+(\text{aq})$ ions, in kJ mol^{-1} ?

- A.** +113.2
- B.** +1.2
- C.** -1.2
- D.** -113.2

(1 mark)